

Error Analysis of a CNN-Based Tomato Leaf Disease Classification Model

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Abstract

This report presents an error analysis of a CNN-based tomato leaf disease classifier covering 10 disease classes. The model was evaluated on 18,160 test images from the PlantVillage dataset, achieving an overall accuracy of 82.26%. However, per-class analysis revealed two systematic weaknesses: (1) Early Blight showed a recall of only 41.5%, meaning the majority of actual Early Blight cases were misclassified, and (2) healthy leaves were frequently confused with Target Spot (218 instances) and Spider Mite damage (143 instances), resulting in a comparatively low recall of 71.8% for the healthy class despite high precision (98.4%). To verify whether this weakness held outside the test dataset, 3 independently-sourced Early Blight images were submitted to the deployed application; all 3 were misclassified into differing wrong classes, corroborating the dataset-level finding. These results indicate specific, actionable weaknesses in the current model that are relevant to its reliability in real-world agricultural use.

1. Introduction

1.1 Background

Automated plant disease detection using deep learning has been widely explored as a way to give farmers access to fast, low-cost diagnostic support, particularly in regions where consulting an agricultural expert is difficult or costly. A CNN-based classifier for this purpose was developed as part of the Farmer Query Solver project, trained on the PlantVillage dataset to identify 10 categories of tomato leaf conditions (9 diseases plus healthy).

1.2 Motivation for This Analysis

An overall accuracy score, while commonly reported, can conceal important failure patterns. A model may perform very well on average while failing consistently on a specific disease, or systematically confusing two particular classes. In an agricultural decision-support context, these hidden weaknesses carry practical consequences: a false disease prediction on a healthy plant may lead to unnecessary fertilizer or pesticide use, while a missed true disease case may result in delayed treatment and crop loss. This report investigates where, specifically, the model's errors occur and considers the practical implications of those patterns.

1.3 Objectives

- To measure per-class precision, recall, and F1-score for the trained model.
- To identify the most significant sources of classification error.
- To independently verify identified weaknesses using images outside the original test set.
- To outline actionable next steps based on the findings.

2. Methodology

2.1 Model and Dataset

The model evaluated is a CNN (`tomato_disease_model.keras`) trained on the PlantVillage tomato subset. The evaluation set consisted of 18,160 labeled images spanning 10 classes.

2.2 Evaluation Procedure

- The model was run on all 18,160 test images.
- Predictions were compared against ground-truth labels to compute a full 10×10 confusion matrix.
- Precision, recall, and F1-score were computed per class.
- Confusion matrix rows for the two weakest-performing classes were examined in detail to characterize the nature of the errors (concentrated vs. scattered).

2.3 Methodological Caveat

The provenance of the 18,160 test images relative to the original training set could not be fully verified — it is possible some overlap exists between the training and test data, which would inflate the reported accuracy relative to true generalization performance. This limitation directly motivated the independent real-world validation described in Section 4.

3. Results

3.1 Overall Performance

Overall accuracy: 82.26% (N = 18,160 images, 10 classes)

3.2 Per-Class Performance

Disease Class	Precision	Recall	F1-Score	Support
Early Blight	0.68	0.42	0.52	1,000
Target Spot	0.71	0.66	0.68	1,404
Spider Mites	0.77	0.75	0.76	1,676
Leaf Mold	0.76	0.78	0.77	952
Septoria Leaf Spot	0.81	0.76	0.78	1,771
Tomato Mosaic Virus	0.86	0.75	0.80	373
Late Blight	0.85	0.85	0.85	1,909
Healthy	0.98	0.72	0.83	1,591
Bacterial Spot	0.78	0.90	0.84	2,127
Yellow Leaf Curl Virus	0.86	0.99	0.92	5,357

Figure 1.

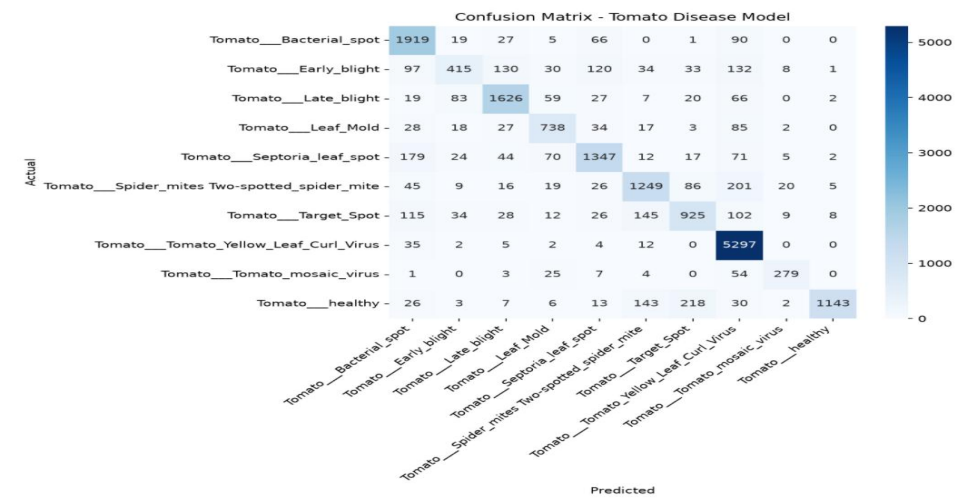


Figure 1: Confusion matrix heatmap for the 10-class tomato disease model.

3.3 Key Findings

Finding 1 — Early Blight recall is low and errors are dispersed. Of 1,000 true Early Blight samples, only 415 (41.5%) were correctly classified. Examining the confusion matrix row for this class shows misclassifications distributed across multiple other classes rather than concentrated in one: 132 → Yellow Leaf Curl Virus, 130 → Late Blight, 120 → Septoria Leaf Spot, 97 → Bacterial Spot.

Finding 2 — Healthy-leaf errors are concentrated in two specific classes. Of 1,591 true healthy samples, 218 (13.7%) were misclassified as Target Spot and 143 (9.0%) as Spider Mites — together accounting for over 75% of all healthy-class errors.

4. Independent Real-World Validation

4.1 Rationale

Given the methodological caveat noted in Section 2.3, an independent check was performed using images not drawn from the test dataset, to assess whether the Early Blight weakness held under real-world conditions.

4.2 Procedure and Results

Three confirmed Early Blight leaf images (sourced independently, verified by filename/origin) were submitted through the deployed application on July 4, 2026.

Figure 2

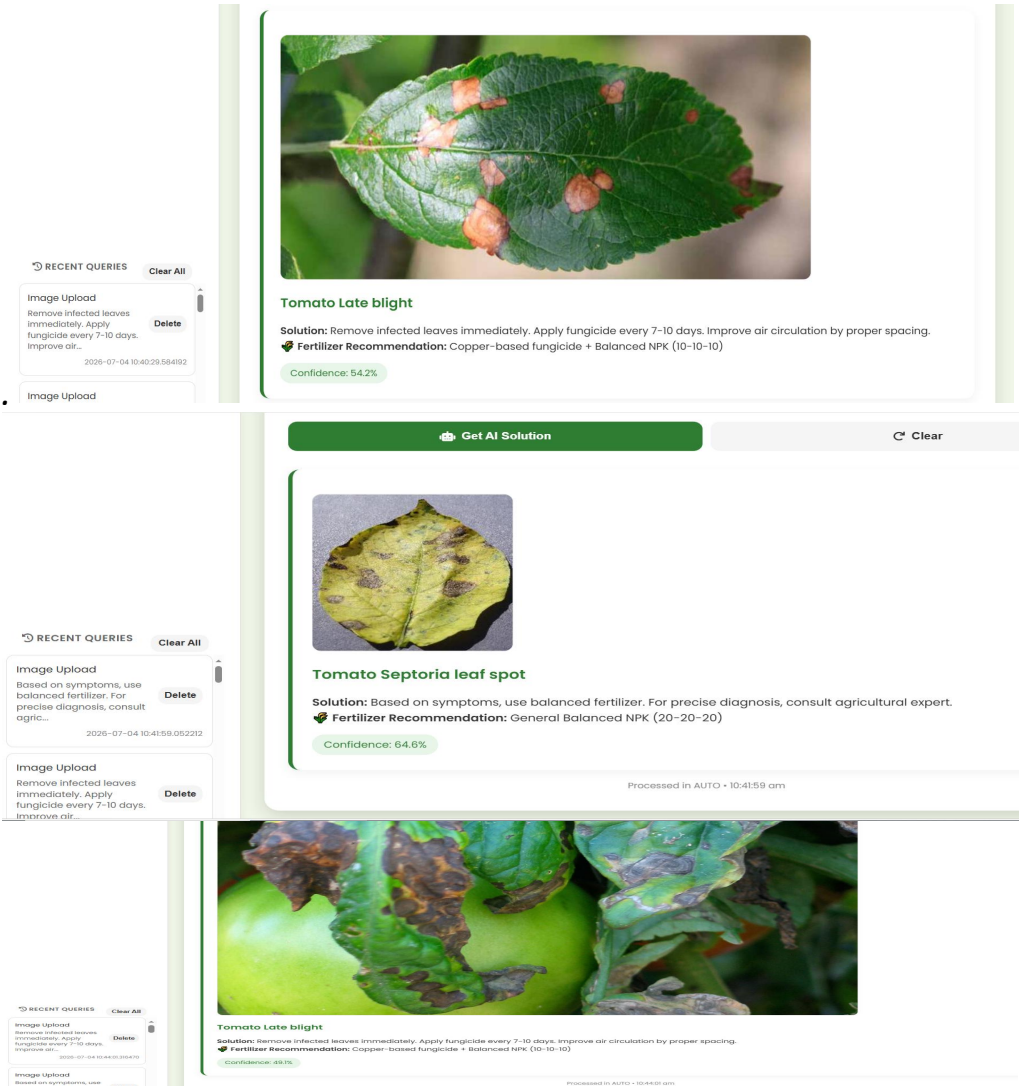


Figure 2: Screenshots of the deployed application's predictions on 3 independently-sourced Early Blight images.

Test	Ground Truth	Model Prediction	Confidence
1	Early Blight	Late Blight (incorrect)	54.2%
2	Early Blight	Septoria Leaf Spot (incorrect)	64.6%
3	Early Blight	Late Blight (incorrect)	49.1%

4.3 Interpretation

All 3 real-world test images were misclassified, and — consistent with Finding 1 — the errors were not concentrated on a single wrong class. This small-sample result ($n=3$) is not statistically conclusive on its own, but its consistency with the large-scale dataset finding strengthens confidence that the Early Blight weakness is a genuine model limitation rather than an artifact of test set construction. The relatively low confidence scores (49–65%) in these predictions, compared to confidence levels above 90% typically observed for correctly-classified Yellow Leaf Curl Virus cases, further suggest the model is comparatively uncertain when processing Early Blight images.

5. Discussion

The two identified weaknesses appear to stem from different underlying causes. The Early Blight recall issue, characterized by dispersed rather than concentrated errors, suggests the model has not learned a strong, distinctive visual representation for this class — plausibly because Early Blight symptoms vary considerably across infection stages, making a single consistent visual signature harder to learn.

The healthy-to-Target-Spot/Spider-Mites confusion pattern is qualitatively different: it is concentrated and repeatable, suggesting the model may be sensitive to minor leaf surface variation (natural blemishes, lighting artifacts) and incorrectly interpreting this as an early disease signal.

From a deployment standpoint, the healthy-leaf misclassification is arguably the more immediately concerning pattern, as it could result in unnecessary treatment recommendations for healthy crops. The Early Blight recall issue is concerning for the opposite reason — genuine disease cases may go undetected.

6. Limitations

- The overlap status between test and training data could not be confirmed, meaning the 82.26% figure may not fully represent generalization to unseen images (Section 2.3).
- All evaluated images (both the 18,160-image test set and the 3 real-world validation images) were relatively clean, well-lit leaf photographs. Performance under realistic field conditions (blur, poor lighting, occlusion, non-standard angles) has not been assessed and is expected to be lower.
- The real-world validation sample ($n=3$) is small; a larger independent sample (e.g., 15–20 images per weak class) would be needed for statistical confidence.
- This analysis identifies weaknesses but does not test corrective interventions (e.g., additional training data, confidence thresholding).

7. Conclusion and Future Work

This analysis shows that an overall accuracy of 82.26% conceals two specific, actionable weaknesses: low recall on Early Blight (41.5%) and a systematic tendency to misclassify healthy leaves as Target Spot or Spider Mite damage. The Early Blight weakness was independently corroborated through real-world testing outside the original dataset.

Recommended next steps:

- Verify test/train data independence to establish a trustworthy accuracy baseline.
- Augment training data for Early Blight, particularly across different infection stages.
- Evaluate model performance on field-realistic (non-curated) images to quantify the dataset-to-real-world accuracy gap.
- Investigate a confidence-threshold mechanism, whereby low-confidence predictions default to an "uncertain — consult a local expert" response rather than a potentially incorrect diagnosis.
- Expand real-world validation to a larger sample size.

References

- Hughes, D.P. and Salathé, M. (2015). An open access repository of images on plant health to enable the development of mobile disease diagnostics. arXiv preprint arXiv:1511.08060 (PlantVillage dataset).
- TensorFlow Documentation, tensorflow.org.

All results reported in this document are derived from an actual evaluation run on July 4, 2026, using the PlantVillage tomato dataset (18,160 images) together with 3 additional independently-sourced real-world test images.